

12 Generalized Linear Models *

12.1 Introduction

In Chapter 10, we discussed logistic regression, which, in the binary case, corresponds to the model $p(y|\mathbf{x}, \mathbf{w}) = \text{Ber}(y|\sigma(\mathbf{w}^\top \mathbf{x}))$. In Chapter 11, we discussed linear regression, which corresponds to the model $p(y|\mathbf{x}, \mathbf{w}) = \mathcal{N}(y|\mathbf{w}^\top \mathbf{x}, \sigma^2)$. These are obviously very similar to each other. In particular, the mean of the output, $\mathbb{E}[y|\mathbf{x}, \mathbf{w}]$, is a linear function of the inputs $\mathbf{x}$ in both cases.

It turns out that there is a broad family of models with this property, known as **generalized linear models** or **GLMs** [MN89].

A GLM is a conditional version of an exponential family distribution (Section 3.4), in which the natural parameters are a linear function of the input. More precisely, the model has the following form:

$$p(y_n|\mathbf{x}_n, \mathbf{w}, \sigma^2) = \exp \left[\frac{y_n \eta_n - A(\eta_n)}{\sigma^2} + \log h(y_n, \sigma^2) \right] \quad (12.1)$$

where $\eta_n \triangleq \mathbf{w}^\top \mathbf{x}_n$ is the (input dependent) natural parameter, $A(\eta_n)$ is the log normalizer, $\mathcal{T}(y_n) = y_n$ is the sufficient statistic, and σ^2 is the dispersion term.¹

We will denote the mapping from the linear inputs to the mean of the output using $\mu_n = \ell^{-1}(\eta_n)$, where the function ℓ is known as the **link function**, and ℓ^{-1} is known as the **mean function**.

Based on the results in Section 3.4.3, we can show that the mean and variance of the response variable are as follows:

$$\mathbb{E}[y_n|\mathbf{x}_n, \mathbf{w}, \sigma^2] = A'(\eta_n) \triangleq \ell^{-1}(\eta_n) \quad (12.2)$$

$$\mathbb{V}[y_n|\mathbf{x}_n, \mathbf{w}, \sigma^2] = A''(\eta_n) \sigma^2 \quad (12.3)$$

12.2 Examples

In this section, we give some examples of widely used GLMs.

1. Technically speaking, GLMs use a slight extension of the natural exponential family known as the **exponential dispersion family**. For a scalar variable, this has the form $p(y|\eta, \sigma^2) = h(y, \sigma^2) \exp \left[\frac{\eta y - A(\eta)}{\sigma^2} \right]$. Here σ^2 is called the **dispersion parameter**. For fixed σ^2 , this is a natural exponential family.

12.2.1 Linear regression

Recall that linear regression has the form

$$p(y_n|\mathbf{x}_n, \mathbf{w}, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_n - \mathbf{w}^\top \mathbf{x}_n)^2\right) \quad (12.4)$$

Hence

$$\log p(y_n|\mathbf{x}_n, \mathbf{w}, \sigma^2) = -\frac{1}{2\sigma^2}(y_n - \eta_n)^2 - \frac{1}{2} \log(2\pi\sigma^2) \quad (12.5)$$

where $\eta_n = \mathbf{w}^\top \mathbf{x}_n$. We can write this in GLM form as follows:

$$\log p(y_n|\mathbf{x}_n, \mathbf{w}, \sigma^2) = \frac{y_n\eta_n - \frac{\eta_n^2}{2}}{\sigma^2} - \frac{1}{2} \left(\frac{y_n^2}{\sigma^2} + \log(2\pi\sigma^2) \right) \quad (12.6)$$

We see that $A(\eta_n) = \eta_n^2/2$ and hence

$$\mathbb{E}[y_n] = \eta_n = \mathbf{w}^\top \mathbf{x}_n \quad (12.7)$$

$$\mathbb{V}[y_n] = \sigma^2 \quad (12.8)$$

12.2.2 Binomial regression

If the response variable is the number of successes in N_n trials, $y_n \in \{0, \dots, N_n\}$, we can use **binomial regression**, which is defined by

$$p(y_n|\mathbf{x}_n, N_n, \mathbf{w}) = \text{Bin}(y_n|\sigma(\mathbf{w}^\top \mathbf{x}_n), N_n) \quad (12.9)$$

We see that binary logistic regression is the special case when $N_n = 1$.

The log pdf is given by

$$\log p(y_n|\mathbf{x}_n, N_n, \mathbf{w}) = y_n \log \mu_n + (N_n - y_n) \log(1 - \mu_n) + \log \binom{N_n}{y_n} \quad (12.10)$$

$$= y_n \log\left(\frac{\mu_n}{1 - \mu_n}\right) + N_n \log(1 - \mu_n) + \log \binom{N_n}{y_n} \quad (12.11)$$

where $\mu_n = \sigma(\eta_n)$. To rewrite this in GLM form, let us define

$$\eta_n \triangleq \log \left[\frac{\mu_n}{(1 - \mu_n)} \right] = \log \left[\frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}_n}} \frac{1 + e^{-\mathbf{w}^\top \mathbf{x}_n}}{e^{-\mathbf{w}^\top \mathbf{x}_n}} \right] = \log \frac{1}{e^{-\mathbf{w}^\top \mathbf{x}_n}} = \mathbf{w}^\top \mathbf{x}_n \quad (12.12)$$

Hence we can write binomial regression in GLM form as follows

$$\log p(y_n|\mathbf{x}_n, N_n, \mathbf{w}) = y_n\eta_n - A(\eta_n) + h(y_n) \quad (12.13)$$

where $h(y_n) = \log \binom{N_n}{y_n}$ and

$$A(\eta_n) = -N_n \log(1 - \mu_n) = N_n \log(1 + e^{\eta_n}) \quad (12.14)$$

Hence

$$\mathbb{E}[y_n] = \frac{dA}{d\eta_n} = \frac{N_n e^{\eta_n}}{1 + e^{\eta_n}} = \frac{N_n}{1 + e^{-\eta_n}} = N_n \mu_n \quad (12.15)$$

and

$$\mathbb{V}[y_n] = \frac{d^2 A}{d\eta_n^2} = N_n \mu_n (1 - \mu_n) \quad (12.16)$$

12.2.3 Poisson regression

If the response variable is an integer count, $y_n \in \{0, 1, \dots\}$, we can use **Poisson regression**, which is defined by

$$p(y_n | \mathbf{x}_n, \mathbf{w}) = \text{Poi}(y_n | \exp(\mathbf{w}^\top \mathbf{x}_n)) \quad (12.17)$$

where

$$\text{Poi}(y | \mu) = e^{-\mu} \frac{\mu^y}{y!} \quad (12.18)$$

is the Poisson distribution. Poisson regression is widely used in bio-statistical applications, where y_n might represent the number of diseases of a given person or place, or the number of reads at a genomic location in a high-throughput sequencing context (see e.g., [Kua+09]).

The log pdf is given by

$$\log p(y_n | \mathbf{x}_n, \mathbf{w}) = y_n \log \mu_n - \mu_n - \log(y_n!) \quad (12.19)$$

where $\mu_n = \exp(\mathbf{w}^\top \mathbf{x}_n)$. Hence in GLM form we have

$$\log p(y_n | \mathbf{x}_n, \mathbf{w}) = y_n \eta_n - A(\eta_n) + h(y_n) \quad (12.20)$$

where $\eta_n = \log(\mu_n) = \mathbf{w}^\top \mathbf{x}_n$, $A(\eta_n) = \mu_n = e^{\eta_n}$, and $h(y_n) = -\log(y_n!)$. Hence

$$\mathbb{E}[y_n] = \frac{dA}{d\eta_n} = e^{\eta_n} = \mu_n \quad (12.21)$$

and

$$\mathbb{V}[y_n] = \frac{d^2 A}{d\eta_n^2} = e^{\eta_n} = \mu_n \quad (12.22)$$

12.3 GLMs with non-canonical link functions

We have seen how the mean parameters of the output distribution are given by $\mu = \ell^{-1}(\eta)$, where the function ℓ is the link function. There are several choices for this function, as we now discuss.

The **canonical link function** ℓ satisfies the property that $\theta = \ell(\mu)$, where θ are the canonical (natural) parameters. Hence

$$\theta = \ell(\mu) = \ell(\ell^{-1}(\eta)) = \eta \quad (12.23)$$

This is what we have assumed so far. For example, for the Bernoulli distribution, the canonical parameter is the log-odds $\theta = \log(\mu/(1 - \mu))$, which is given by the logit transform

$$\theta = \ell(\mu) = \text{logit}(\mu) = \log\left(\frac{\mu}{1 - \mu}\right) \quad (12.24)$$

The inverse of this is the sigmoid or logistic function $\mu = \sigma(\theta) = 1/(1 + e^{-\theta})$.

However, we are free to use other kinds of link function. For example, the **probit link function** has the form

$$\eta = \ell(\mu) = \Phi^{-1}(\mu) \quad (12.25)$$

Another link function that is sometimes used for binary responses is the **complementary log-log** function

$$\eta = \ell(\mu) = \log(-\log(1 - \mu)) \quad (12.26)$$

This is used in applications where we either observe 0 events (denoted by $y = 0$) or one or more (denoted by $y = 1$), where events are assumed to be governed by a Poisson distribution with rate λ . Let E be the number of events. The Poisson assumption means $p(E = 0) = \exp(-\lambda)$ and hence

$$p(y = 0) = (1 - \mu) = p(E = 0) = \exp(-\lambda) \quad (12.27)$$

Thus $\lambda = -\log(1 - \mu)$. When λ is a function of covariates, we need to ensure it is positive, so we use $\lambda = e^\eta$, and hence

$$\eta = \log(\lambda) = \log(-\log(1 - \mu)) \quad (12.28)$$

12.4 Maximum likelihood estimation

GLMs can be fit using similar methods to those that we used to fit logistic regression. In particular, the negative log-likelihood has the following form (ignoring constant terms):

$$\text{NLL}(\mathbf{w}) = -\log p(\mathcal{D}|\mathbf{w}) = -\frac{1}{\sigma^2} \sum_{n=1}^N \ell_n \quad (12.29)$$

where

$$\ell_n \triangleq \eta_n y_n - A(\eta_n) \quad (12.30)$$

where $\eta_n = \mathbf{w}^\top \mathbf{x}_n$. For notational simplicity, we will assume $\sigma^2 = 1$.

We can compute the gradient for a single term as follows:

$$\mathbf{g}_n \triangleq \frac{\partial \ell_n}{\partial \mathbf{w}} = \frac{\partial \ell_n}{\partial \eta_n} \frac{\partial \eta_n}{\partial \mathbf{w}} = (y_n - A'(\eta_n)) \mathbf{x}_n = (y_n - \mu_n) \mathbf{x}_n \quad (12.31)$$

where $\mu_n = f(\mathbf{w}^\top \mathbf{x}_n)$, and f is the inverse link function that maps from canonical parameters to mean parameters. For example, in the case of logistic regression, $f(\eta_n) = \sigma(\eta_n)$, so we recover

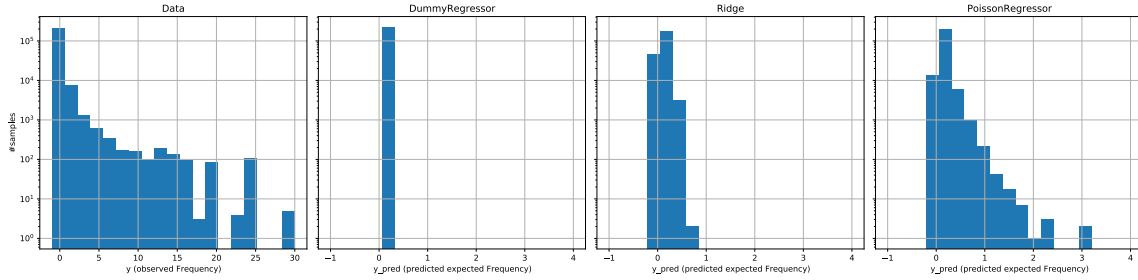


Figure 12.1: Predictions of insurance claim rates on the test set. (a) Data. (b) Constant predictor. (c) Linear regression. (d) Poisson regression. Generated by `poisson_regression_insurance.ipynb`.

Equation (10.21). This gradient expression can be used inside SGD, or some other gradient method, in the obvious way.

The Hessian is given by

$$\mathbf{H} = \frac{\partial^2}{\partial \mathbf{w} \partial \mathbf{w}^\top} \text{NLL}(\mathbf{w}) = - \sum_{n=1}^N \frac{\partial \mathbf{g}_n}{\partial \mathbf{w}^\top} \quad (12.32)$$

where

$$\frac{\partial \mathbf{g}_n}{\partial \mathbf{w}^\top} = \frac{\partial \mathbf{g}_n}{\partial \mu_n} \frac{\partial \mu_n}{\partial \mathbf{w}^\top} = -\mathbf{x}_n f'(\mathbf{w}^\top \mathbf{x}_n) \mathbf{x}_n^\top \quad (12.33)$$

Hence

$$\mathbf{H} = \sum_{n=1}^N f'(\eta_n) \mathbf{x}_n \mathbf{x}_n^\top \quad (12.34)$$

For example, in the case of logistic regression, $f(\eta_n) = \sigma(\eta_n)$, and $f'(\eta_n) = \sigma(\eta_n)(1 - \sigma(\eta_n))$, so we recover Equation (10.23). In general, we see that the Hessian is positive definite, since $f'(\eta_n) > 0$; hence the negative log likelihood is convex, so the MLE for a GLM is unique (assuming $f(\eta_n) > 0$ for all n).

Based on the above results, we can fit GLMs using gradient based solvers in a manner that is very similar to how we fit logistic regression models.

12.5 Worked example: predicting insurance claims

In this section, we give an example of predicting insurance claims using linear and Poisson regression.² The goal is to predict the expected number of insurance claims per year following car accidents. The dataset consists of 678k examples with 9 features, such as driver age, vehicle age, vehicle power,

2. This example is from https://scikit-learn.org/stable/auto_examples/linear_model/plot_poisson_regression_non_normal_loss.html

Name	MSE	MAE	Deviance
Dummy	0.564	0.189	0.625
Ridge	0.560	0.177	0.601
Poisson	0.560	0.186	0.594

Table 12.1: Performance metrics on the test set. MSE = mean squared error. MAE = mean absolute error. Deviance = Poisson deviance.

etc. The target is the frequency of claims, which is the number of claims per policy divided by the exposure (i.e., the duration of the policy in years).

We plot the test set in Figure 12.1(a). We see that for 94% of the policies, no claims are made, so the data has lots of 0s, as is typical for count and rate data. The average frequency of claims is 10%. This can be converted into a dummy model, which always predicts this constant. This results in the predictions shown in Figure 12.1(b). The goal is to do better than this.

A simple approach is to use linear regression, combined with some simple feature engineering (binning the continuous values, and one-hot encoding the categorical). (We use a small amount of ℓ_2 regularization, so technically this is ridge regression.) This gives the results shown in Figure 12.1(c). This is better than the baseline, but still not very good. In particular, it can predict negative outcomes, and fails to capture the long tail.

We can do better using Poisson regression, using the same features but a log link function. The results are shown in Figure 12.1(d). We see that predictions are much better.

An interesting question is how to quantify performance in this kind of problem. If we use mean squared error, or mean absolute error, we may conclude from Table 12.1 that ridge regression is better than Poisson regression, but this is clearly not true, as shown in Figure 12.1. Instead it is more common to measure performance using the **deviance**, which is defined as

$$D(\mathbf{y}, \hat{\boldsymbol{\mu}}) = 2 \sum_i (\log p(y_i | \mu_i^*) - \log p(y_i | \mu_i)) \quad (12.35)$$

where μ_i is the predicted parameters for the i 'th example (based on the input features $\mathbf{x}_i$ and the training set $\mathcal{D}$), and μ_i^* is the optimal parameter estimated by fitting the model just to the true output y_i . (This is the so-called saturated model, that perfectly fits the test set.) In the case of Poisson regression, we have $\mu_i^* = y_i$. Hence

$$D(\mathbf{y}, \boldsymbol{\mu}) = 2 \sum_i [(y_i \log y_i - y_i - \log(y_i!)) - (y_i \log \hat{\mu}_i - \hat{\mu}_i - \log(y_i!))] \quad (12.36)$$

$$= 2 \sum_i \left[(y_i \log \frac{y_i}{\hat{\mu}_i} + \hat{\mu}_i - y_i) \right] \quad (12.37)$$

By this metric, the Poisson model is clearly better (see last column of Table 12.1).

We can also compute a **calibration plot**, which plots the actual frequency vs the predicted frequency. To compute this, we bin the predictions into intervals, and then count the empirical frequency of claims for all examples whose predicted frequency falls into that bin. The results are shown in Figure 12.2. We see that the constant baseline is well calibrated, but of course it is not very accurate. The ridge model is miscalibrated in the low frequency regime. In particular, it

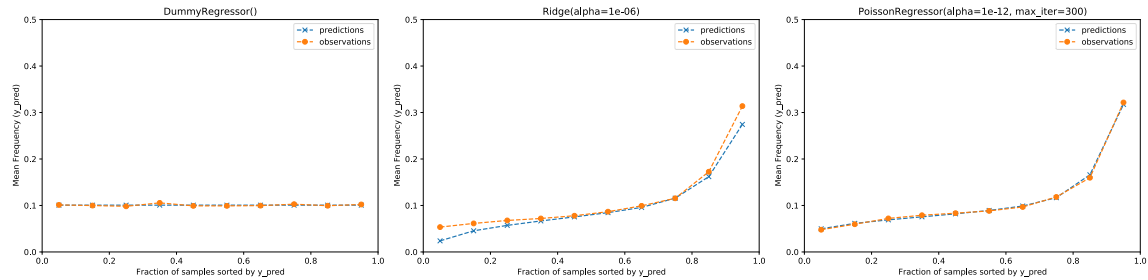


Figure 12.2: Calibration plot for insurance claims prediction. Generated by `poisson_regression_insurance.ipynb`.

underestimates the total number of claims in the test set to be 10,693, whereas the truth is 11,935. The Poisson model is better calibrated (i.e., when it predicts examples will have a high claim rate, they do in fact have a high claim rate), and it predicts the total number of claims to be 11,930.